Dong et al. 2020 — Prostate Cancer · CCCA cell-state re-analysis

Prostate adenocarcinoma + NEPC · 10x · 9,698 malignant cells (4 samples)

GO / NO-GO GATES

Gate 1 — mixing & continuity

GO

One connected kNN component; adeno (#1/#2/#5) intermix and are continuous with the NEPC pole (#6). iLISI rises 1.00 → 1.70 (ceiling 4) under Harmony θ=10, plateauing by θ=20; n_pcs=30 (robust band). Overturns the prior scalar-led NO-GO (self-neighbour enrichment) — the embedding is one continuum.

Gate 2 — module geography

GO (non-branching)

Single developmental axis: Gast & pSom co-vary (r=+0.91) at one pole; Crest inverse-gradient terminus. Crest opposes Gast (−0.34) and pSom (−0.48); joint pSom&Crest enrichment 0.66 (<1 → inverse, not a hub). No Gast arm exists to invert onto → no Crest-arm flag. Passes as the non-branching case.

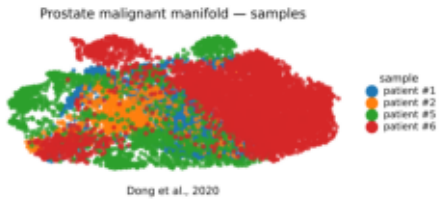
Gate 3 — hub cross-repression

NA

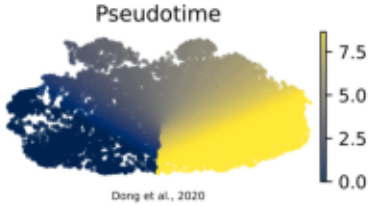
Committed principal curve is 2 tips / 0 forks. Branching check (shown, not asserted): a forced PPT tree (Nodes=200) gives 7 tips / 5 branch nodes — but these are node-density over-segmentation, not a bifurcation: branch count collapses 5→4→3→2→1 as PPT nodes drop 200→100→50→30→20, and the 7 tips co-vary in Gast & pSom (r=+0.90), not competing arms. No bifurcation → Gate 3 not applicable.

STEP FIGURES

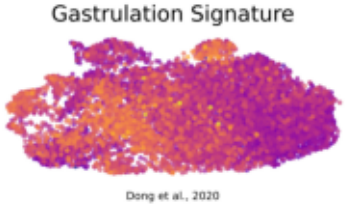
01 · samples (Gate 1)



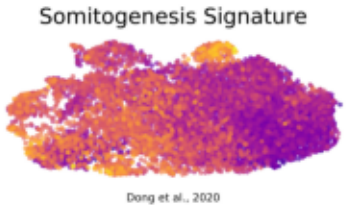
02 · pseudotime



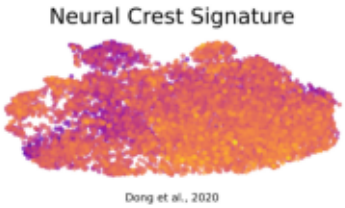
03 · Gastrulation



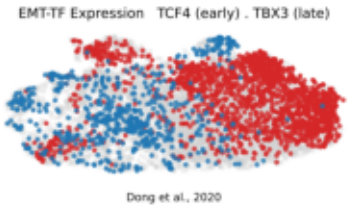
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TF (TCF4→TBX3)



Dataset. Primary prostate tumours; 5 adenocarcinoma + 1 small-cell NEPC. Malignant compartment, 9,698 cells across 4 malignant-bearing samples. Expression: log1p of normalized UMI counts.

Verdict. The pattern reproduces as the NON-BRANCHING case — one gastrulation/somitogenesis-high origin (adeno) → neural-crest-high terminus (NEPC) axis, cross-repression collapsed. Geometry: Linear.

KEY PARAMETER CHOICES

Min-cell threshold
floor ≥100 (gap premise fails: counts 915–4765, gap at top)
n_pcs / θ
30 (robust 20–40 band) / Harmony θ=10 (iLISI plateau)
Geometry
Linear — principal curve (ElPiGraph, 20 nodes, fold-free 2.64)
Module recovery
Gast 71/81 · pSom 83/98 · Crest 99/104
Gast μ / pSom μ / Crest μ
+0.035 / +0.100 / −0.041 (raw score_genes means)

Samples kept
4/4 malignant-bearing (of 6 total patients)
Overdispersed genes
1,670
Root
Gast/pSom-high curve tip (developmental origin, Crest-low)
EMT-TFs (derived)
TCF4 (early/root) → TBX3 (late/terminus), detection-gated
Crest terminus
Crest-high decile mean pseudotime 0.92 (terminal)

Hub identity: NA (non-branching, no convergence hub) · Hub r: NA · Gate ledger: GO / GO / NA